Author Response 1

Reviewer: 1

Comments to the Author

Though articles included in this paper were heterogeneous, it still has some referenced value. This result of this systematic review and meta-analysis may approach the real status of convalescent plasma therapy. But the author did not explain how to exclude the influence of antiviral drugs, and I did not find the excluded reason of forty-two studies.

R: *Thank you for your valuable comments. We have addressed your suggestion about the influence of antiviral drugs in the discussion section of our manuscript, as follows:*

Regarding the influence of antiviral drugs used in the articles included in this systematic review, the study by Li19 reported the use of antiviral drugs, steroids, immunoglobulin, and interferon, since they have treated patients with severe or life-threatening disease. For its part, the Agarwal study reported the joint use of plasma with some of these drugs: hydroxychloroquine, remdesivir, lopinavir/ritonavir, oseltamivir, steroids, and tocilizumab18. The Abolghasemi study reported the concomitant use of Lopinavir/Ritonavir, Hydroxychloroquine, and an anti-inflammatory agent16 together with plasma therapy. Previous studies report some benefits from using these drugs in adult patients37,38; however, more studies are needed here to control this variable, although, in clinical practice, this is almost impossible when treating patients with severe disease. The only study that could control this variable was Libster17 because it only included patients with early-stage disease.

Please, note that the complete list of excluded studies, with their reason is available in the supplemental file:

Table 2. Excluded studies with reasons

Reference	Author	Reason for excluding
1	Salazar	single arm
2	Wang	no control group
3	Skendros	in vitro study
4	Choudhuru	protocol
5	Olivares-Gazca	no control group
6	Teofili	protocol
7	Bradfute	single arm
8	Mayer	data science study
9	Mo	not trial
10	Rodríguez-Rubio	protocol
11	Ragnesola	single arm
12	Johansson	another drug
13	Li	duplicated
14	Agarwal	duplicated
15	Kumar	not trial
16	Omrani	no control group
17	Moore	case report
18	Khamis	case series
19	Meo	not trial
20	Sruwerdt	not trial
21	Moniuszko-Malinowska	case report
22	Ibrahim	single arm
23	Liu	retrospective
24	González	Observational
25	Liu	case report
13	Li	duplicated
26	Saverino	not trial
27	Chen	not trial
28	Korper	protocol
29	Bobek	case report
30	Perotti	protocol
31	Shen	case report
32	Duan	case series
14	Agarwal	duplicated
33	Devos	protocol
34	Lou	another drug
35	Salazar	not trial

Reviewer: 2

Comments to the Author

I don't think that there is need of including supplementary files in the review article document this much detail like the list of excluded references.

R: *Thank you for your valuable feed-back and suggestions. In this case we decided to include the list of the excluded references for replication purposes, if somebody needs to consult it, it is available. Also, Reviewer 1 asked for some details contained in the supplemental file.*

Reviewer: 3

Comments to the Author

Abeldano et al performed a meta-analysis of the trials addressing convalescent plasma in hospitalized patients with covid-19. They conclude that convalescent plasma cannot reduce the mortality or ventilation requirement in hospitalized patients with covid-19.

Major comments

Trial selection and randomization

The trials that were assessed are displayed in the electronical supplement.

The Simonovich trial (A Randomized Trial of Convalescent Plasma in Covid-19 Severe Pneumonia, N Engl J Med Nov 2020) is not referred to. https://pubmed.ncbi.nlm.nih.gov/33232588/

R*: Thank you for your suggestion. The Simonovich trial was not included at the time of the first searching because it was published on feb 2021. Randomized Controlled Trial N Engl J Med. 2021 Feb 18;384(7):619-629.*

In the current version of our manuscript, it was included. The results have been included this article in the qualitative synthesis and in the meta-analyses.

The Gharbharan trial (Convalescent Plasma for COVID-19. A randomized clinical trial. medRxiv 2020.07.01.20139857; doi: https://doi.org/10.1101/2020.07.01.20139857, prematurely interrupted) should be discussed.

R: *Yes, it was included in the discussion section, supporting the early-stage plasma therapy, as follows:*

This early-stage plasma therapy is also supported by a study conducted in the Netherlands36, which had to be halted prematurely. The authors showed that late-administered plasma therapy did not benefit patients because they already had high virus-neutralizing antibody titers on the day of study inclusion with titers comparable to the recovered donors.

The Abolghasami trial has been selected by the authors but is not randomized. Interestingly, it is unclear whether the authors consider it randomized (page 8, line 23), or not (table 1). The issue of randomization in the selected trials should be addressed seriously as it is one of the most important way to reduce selection biases (which is obvious in the Abolghasami trial) and favor group comparability (which is difficult in hospitalized COVID-19 patient).

R: *Yes, you are right, the Abolghasemi trial was non-randomized. The mistake was corrected, we replaced the word "randomization" by "allocation" in the manuscript:*

(…) and three studies reported mortality at any time from allocation [16,17, 22]

Additional comments

The conclusion should not be so definitive, in particular because of the flaws in the methodologies of the selected trials.

R: *Conclusion was amended to be not so definitive, as follows:*

The current evidence suggests that transfusion with convalescent plasma cannot reduce the mortality or ventilation requirement in hospitalized patients diagnosed with SARS-CoV-2 infection. More controlled clinical trials conducted with methodologies that ensure a low risk of bias are still needed.

These flaws (risk of bias) should be displayed in the result section of the abstract (not only in the conclusion).

R: It was included in the results section of the abstract:

The overall risk of bias in the reviewed articles was established at low-risk only in two randomized, double-blind clinical trial.

A description of the quality of the plasma used, in terms of neutralizing antibodies against SARScoV-2 should be provided.

R: It was included in table 1, including IgG titer concentration for each study.